



LAB ASSESSMENT 2 : JENKINS

NAME: SHARU

SREE.S

REG NO.

:23MIS0403

COURSE CODE: ISWE406P

SLOT : L03+L04

```
[CRLF]> *****  
[CRLF]> *****  
[CRLF]> *****  
[CRLF]> *****  
[CRLF]> Jenkins initial setup is required. An admin user has been created and a password generated.  
[CRLF]> Please use the following password to proceed to installation:  
[CRLF]>  
[CRLF]> 79af67ef86ba4bee914a29f65eacb858  
[CRLF]>  
[CRLF]> This may also be found at: C:\Users\likit\.jenkins\secrets\initialAdminPassword  
[CRLF]>  
[CRLF]> *****  
[CRLF]> *****  
[CRLF]> *****  
  
2026-02-04 11:16:43.030+0000 [id=62] INFO jenkins.InitReactorRunner$1onAttained: Completed initialization  
2026-02-04 11:16:43.077+0000 [id=61] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
```

Getting Started

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log (not sure where to find it?) and this file on the server:

`C:\Users\likit\.jenkins\secrets\initialAdminPassword`

Please copy the password from either location and paste it below.

Administrator password

.....

Continue

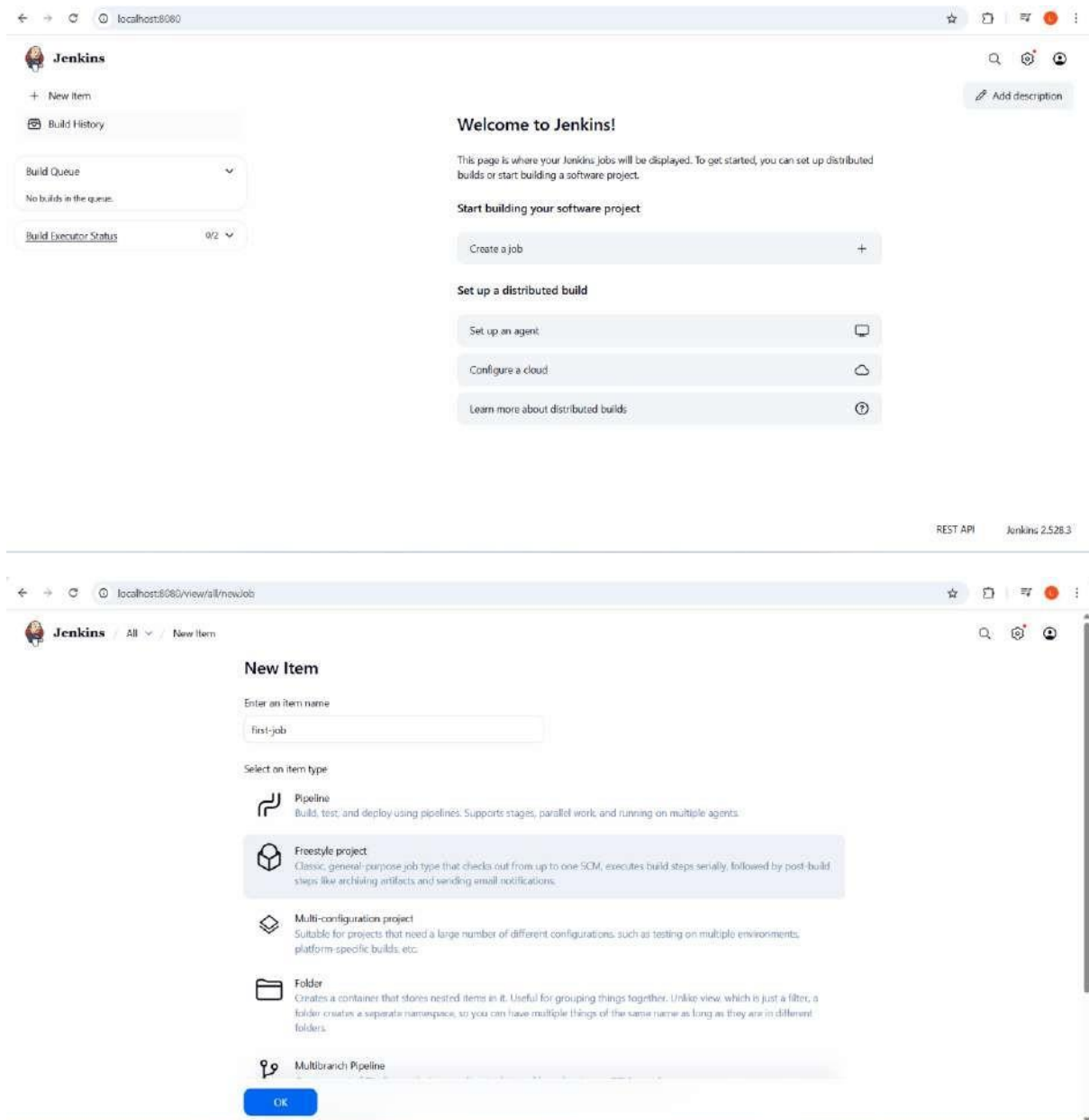
Getting Started

Getting Started

☒ Folders	☒ OWASP Markup Formatter	☐ Build Timeout	☐ Credentials Binding	** commons-lang3 v3.x Jenkins API ** Tonicons API Folders OWASP Markup Formatter ** ASM API ** JSON Path API ** Struts ** Pipeline: Step API
☐ Timestampers	☐ Workspace Cleanup	☐ Ant	☐ Gradle	
☐ Pipeline	☐ GitHub Branch Source	☐ Pipeline: GitHub Groovy Libraries	☐ Pipeline Graph View	
☐ Git	☐ SSH Build Agents	☐ Matrix Authorization Strategy	☐ LDAP	
☐ Email Extension	☐ Mailer	☐ Dark Theme		
				** - required dependency

Jenkins 2.528.3

3 &4 Jenkins UI overview & Create **Freestyle Job**



5 Add simple build step:

- Windows: echo Hello Jenkins

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute Windows batch command ?

Command

See the list of available environment variables

echo Hello Jenkins

Advanced ▾

+ Add build step

Post-build Actions

Save

Apply

6.Run job manually

localhost:8080/job/first-job/

Jenkins / first-job

Status

</> Changes

Workspace

Build Now

Configure

Delete Project

Rename

Builds

Today

✓ #1 5:32 PM

first-job

first jenkins task

Permalinks

Jenkins / first-job

Status

</> Changes

Workspace

Build Now

Configure

Delete Project

Rename

Builds

Today

✓ #1 5:32 PM

first-job

first jenkins task

Permalinks

Build scheduled

localhost:8080

7. View:

- Console Output

○ Build History

 **Jenkins** / first-job

Status

</> Changes

Workspace

Build Now

Configure

Delete Project

Rename

first-job

first jenkins task

Permalinks

Builds

Today

✓ #1 5:32 PM

</> Changes

Console Output

Edit Build Information

Delete build '#1'

Timings

 **Jenkins** / first-job / #1 / Console Output

Status

</> Changes

Console Output

Edit Build Information

Delete build '#1'

Timings

✓ Console Output

Download

Copy

View as plain text

Started by user P Likhitha

Running as SYSTEM

Building in workspace C:\Users\likit\.jenkins\workspace\first-job

[First-job] \$ cmd /c call c:\Users\likit\AppData\Local\Temp\jenkins13900017057337067216.bat

c:\Users\likit\.jenkins\workspace\first-job>echo Hello Jenkins

Hello Jenkins

C:\Users\likit\.jenkins\workspace\first-job>exit 0

Finished: SUCCESS

Jenkins / first-job / #1

Status

Changes

Console Output

Edit Build Information

Delete build '#1'

Timings

#1 (Feb 4, 2026, 5:32:44 PM)

Started by user P.Likitha

This run spent:

- 0.11 sec waiting
- 0.8 sec build duration
- 0.91 sec total from scheduled to completion

No changes

Add description

Keep this build forever

Started 8 min 39 sec ago
Took 0.8 sec

Jenkins / first-job

Status

Changes

Workspace

Build Now

Configure

Delete Project

Rename

first-jobs

first-jenkins task

Permalinks

- Last build (#1), 17 min ago
- Last stable build (#1), 17 min ago
- Last successful build (#1), 17 min ago
- Last completed build (#1), 17 min ago

Builds

Filter

Today

#1 5:32 PM

Jenkins / All / Build History

+ New Item

Build History

Build Queue

No builds in the queue.

Build Executor Status

0/2

Build History of Jenkins

S	Build	Time Since	Status
✓	first-job #1	19 min	stable

Icons: S M L

II. Learning Goal

Create GitHub repository

1
General

Owner *

Repository name *

Liki-git

jenkins-demo

jenkins-demo is available.

Great repository names are short and memorable. How about [solid-doodle?](#)

Description

0 / 350 characters

2
Configuration

Choose visibility *

Choose who can see and commit to this repository

Public

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

On

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

No .gitignore

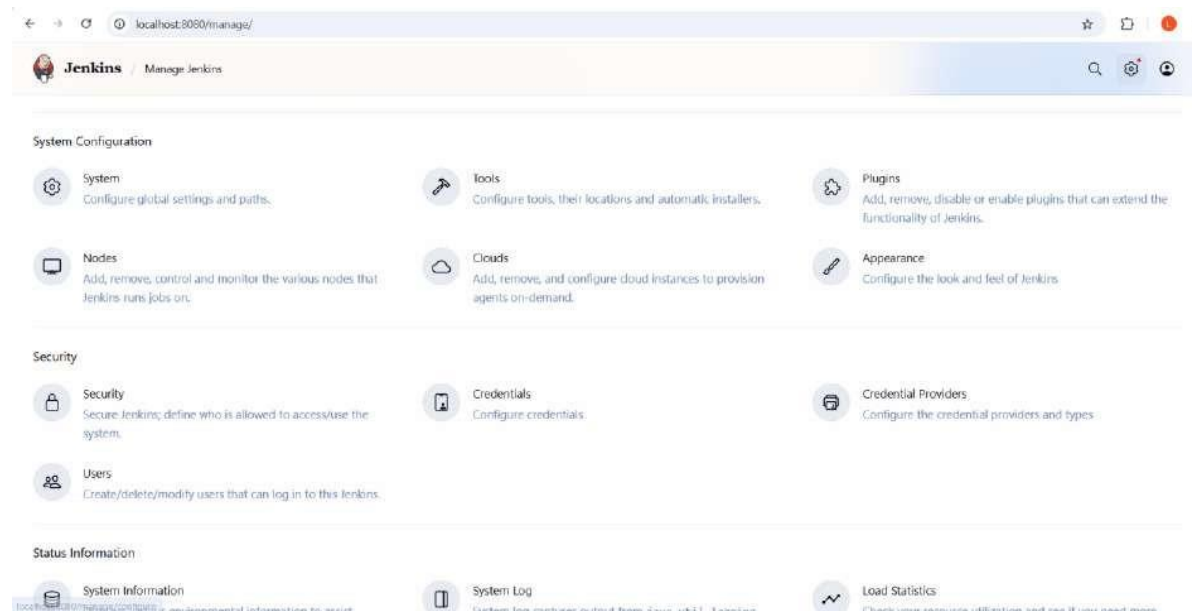
Add license

Licenses explain how others can use your code. [About licenses](#)

No license

Create repository

Configure Git in Jenkins



Jenkins / Manage Jenkins / Tools

Git installations

Git

Name
Default

Path to Git executable
C:\Program Files\Git\bin\git.exe

☐ Install automatically

+ Add Git

Gradle installations

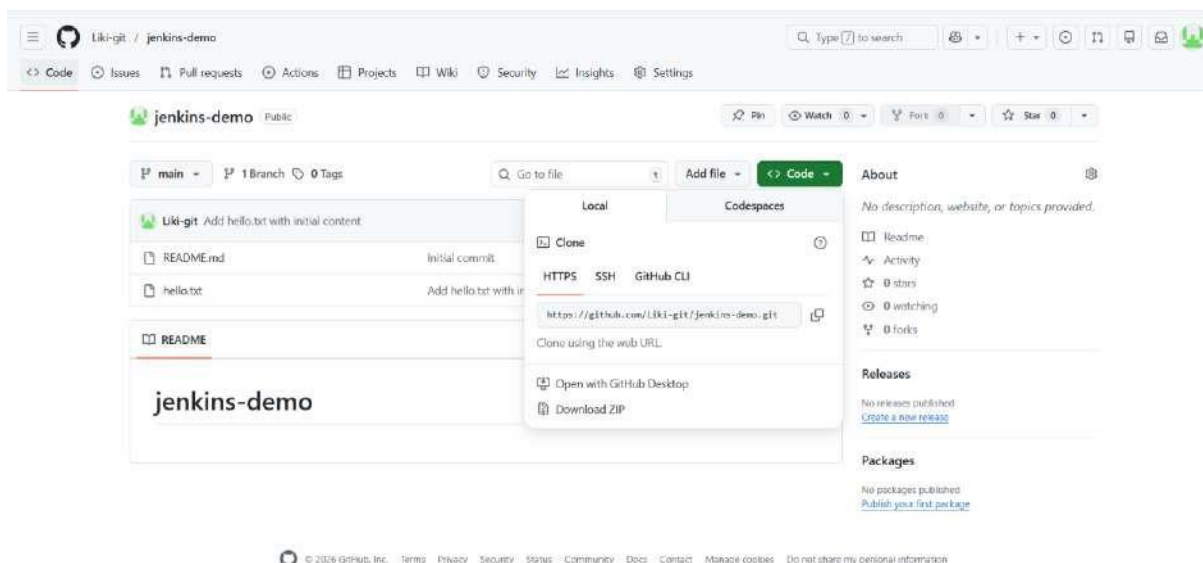
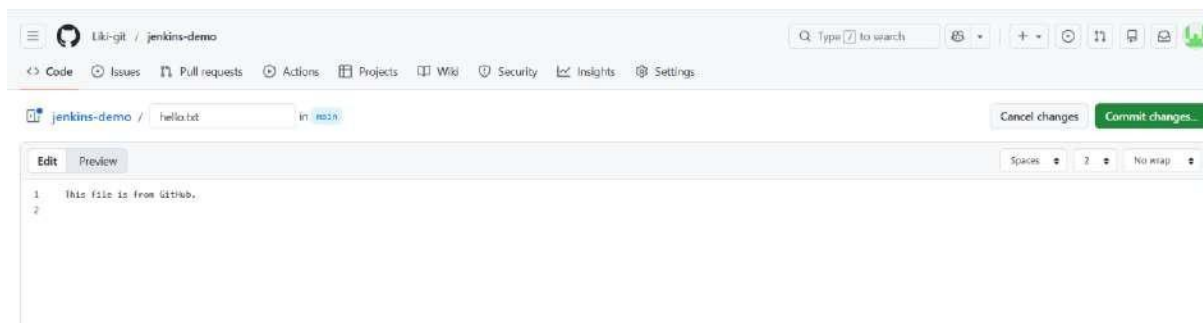
+ Add Gradle

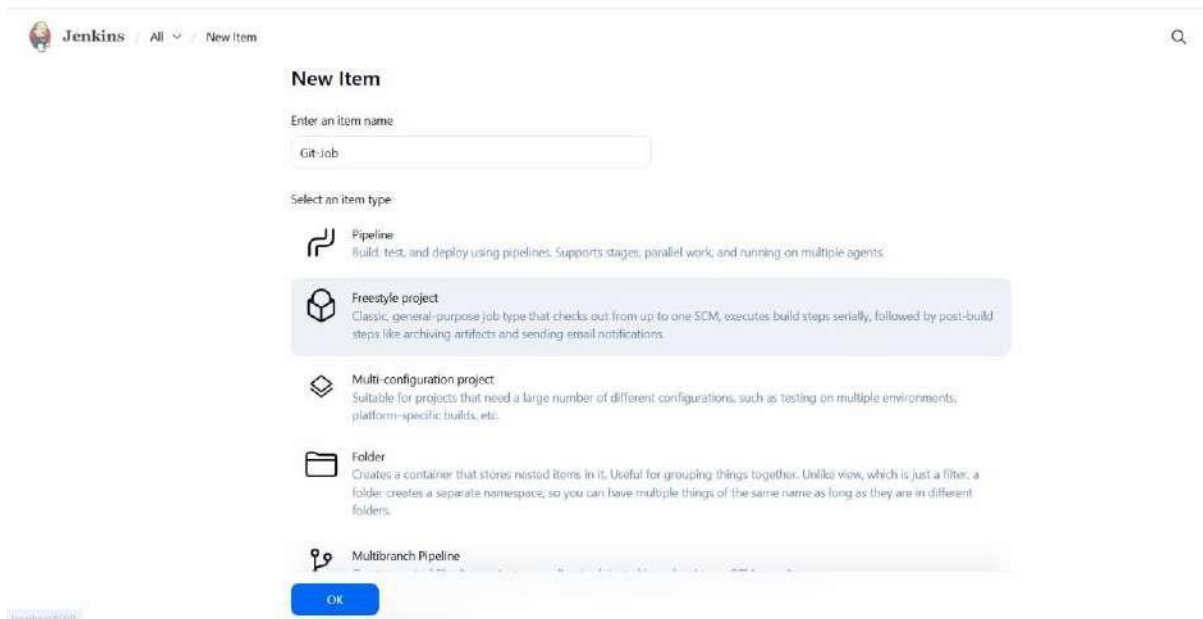
Save

Apply

Create Freestyle Job:

- Source Code Management → Git





The Jenkins 'New Item' page is shown. At the top, there's a header with the Jenkins logo, 'All' dropdown, and 'New Item' link. A search icon is on the right. The main heading is 'New Item'. Below it, there's a text input field for 'Enter an item name' with 'Git-Job' entered. Then, a section 'Select an item type:' lists several options: 'Pipeline' (Build, test, and deploy using pipelines...), 'Freestyle project' (Classic, general-purpose job type...), 'Multi-configuration project' (Suitable for projects that need a large number of different configurations...), 'Folder' (Creates a container that stores nested Items...), and 'Multibranch Pipeline'. An 'OK' button is at the bottom right.

Jenkins / All / New Item

New Item

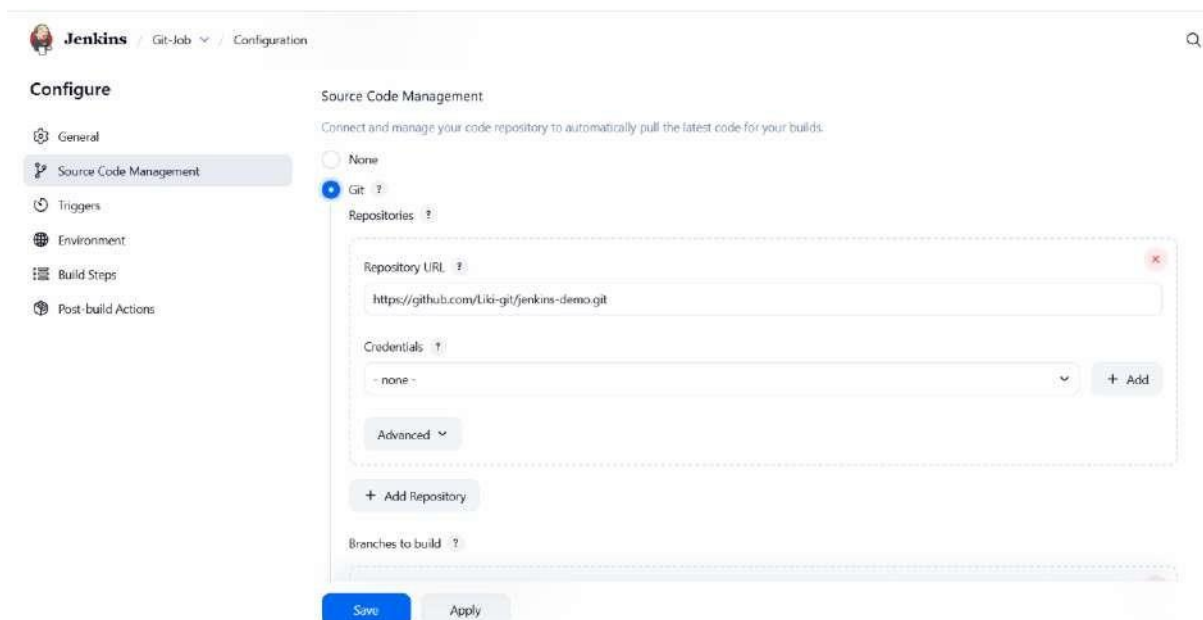
Enter an item name

Git-Job

Select an item type:

- Pipeline**
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
- Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Multi-configuration project**
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
- Folder**
Creates a container that stores nested Items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
- Multibranch Pipeline**

OK



The Jenkins 'Configuration' page for 'Git-Job' is shown. The left sidebar has a 'Configure' heading and a list of tabs: 'General', 'Source Code Management' (selected), 'Triggers', 'Environment', 'Build Steps', and 'Post-build Actions'. The main content area is titled 'Source Code Management' and has a description: 'Connect and manage your code repository to automatically pull the latest code for your builds.' There are radio buttons for 'None' and 'Git' (selected). Below, there's a 'Repositories' section with a 'Repository URL' field containing 'https://github.com/Liud-git/jenkins-demo.git', a 'Credentials' dropdown set to 'none', and an 'Add' button. There's also an 'Advanced' dropdown and an '+ Add Repository' button. At the bottom, there's a 'Branches to build' section and 'Save' and 'Apply' buttons.

Jenkins / Git-Job / Configuration

Configure

- General
- Source Code Management**
- Triggers
- Environment
- Build Steps
- Post-build Actions

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ Git

Repositories

Repository URL:

Credentials: + Add

Advanced

+ Add Repository

Branches to build

Save Apply

Add build step:

- Windows batch / Shell

 **Jenkins** / Git-Job / Configuration

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

☐ Add timestamps to the Console Output

☐ Inspect build log for published build scans

☐ Terminate a build if it's stuck

☐ With Ant ?

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute Windows batch command ?

Command

See the list of available environment variables

echo Code pulled successfully
git

Advanced ▾

+ Add build step

Save

Apply

 **Jenkins** / Git-Job

Status

</> Changes

Workspace

Build Now

Configure

Delete Project

Rename

Builds

Filter

Today

✓ #2 6:13 PM

localhost:8080/job/Git-Job/2/console

Jenkins Git-Job #2 Console Output

Download Copy View as plain text

Console Output

```
Started by user P Likitha
Running as SYSTEM
Building in workspace C:\Users\likit\.jenkins\workspace\Git-Job
The recommended git tool is: NONE
No credentials specified
> C:\Program Files\Git\bin\git.exe rev-parse --resolve-git-dir C:\Users\likit\.jenkins\workspace\Git-Job\.git # timeout=10
Fetching changes from the remote Git repository
> C:\Program Files\Git\bin\git.exe config remote.origin.url https://github.com/Liki-git/jenkins-demo.git # timeout=10
Fetching upstream changes from https://github.com/Liki-git/jenkins-demo.git
> C:\Program Files\Git\bin\git.exe --version # timeout=10
> git --version # 'git version 2.52.0.windows.1'
> C:\Program Files\Git\bin\git.exe fetch --tags --force --progress -- https://github.com/Liki-git/jenkins-demo.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
> C:\Program Files\Git\bin\git.exe rev-parse "refs/remotes/origin/main"{commit}" # timeout=10
Checking out Revision 410fadc29973be777003f1e1e85673b8e2b95e6f (refs/remotes/origin/main)
> C:\Program Files\Git\bin\git.exe config core.sparsecheckout # timeout=10
> C:\Program Files\Git\bin\git.exe checkout -f 410fadc29973be777003f1e1e85673b8e2b95e6f # timeout=10
Commit message: "Add hello.txt with initial content"
First time build. Skipping changelog.
[Git-Job] $ cmd /c call C:\Users\likit\AppData\Local\Temp\jenkins8968919513663670638.bat

C:\Users\likit\.jenkins\workspace\Git-Job>echo code pulled successfully
code pulled successfully

C:\Users\likit\.jenkins\workspace\Git-Job>dir
```

Jenkins Git-Job #2 Console Output

Download Copy View as plain text

```
> C:\Program Files\Git\bin\git.exe checkout -f 410fadc29973be777003f1e1e85673b8e2b95e6f # timeout=10
Commit message: "Add hello.txt with initial content"
First time build. Skipping changelog.
[Git-Job] $ cmd /c call C:\Users\likit\AppData\Local\Temp\jenkins8968919513663670638.bat

C:\Users\likit\.jenkins\workspace\Git-Job>echo code pulled successfully
code pulled successfully

C:\Users\likit\.jenkins\workspace\Git-Job>dir
Volume in drive C is windows-SSD
Volume Serial Number is AC5E-ED6E

Directory of C:\Users\likit\.jenkins\workspace\Git-Job

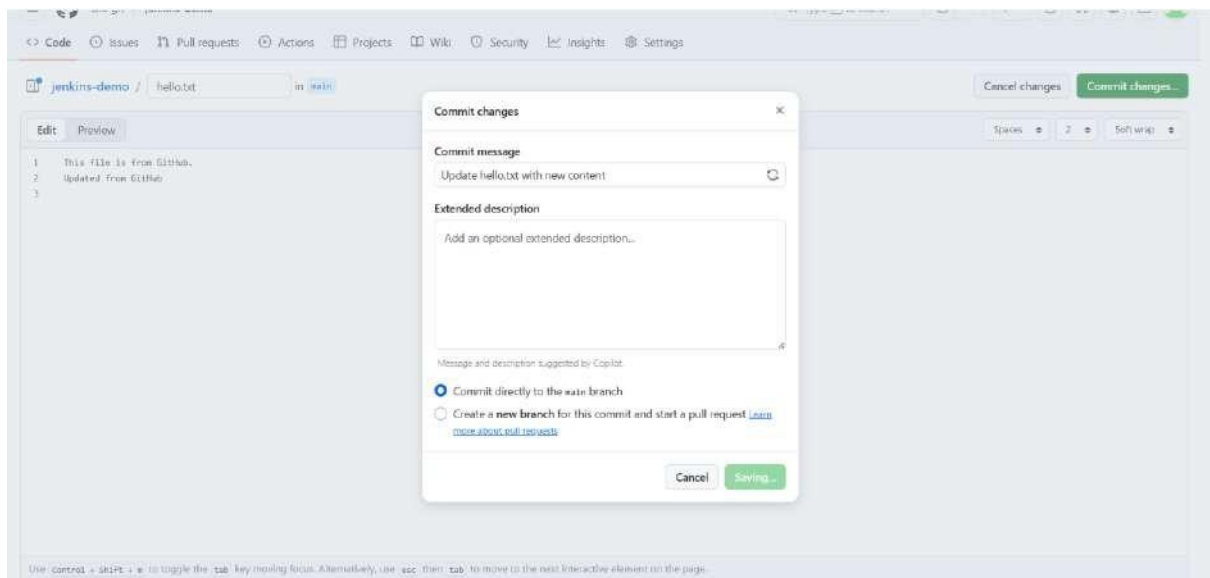
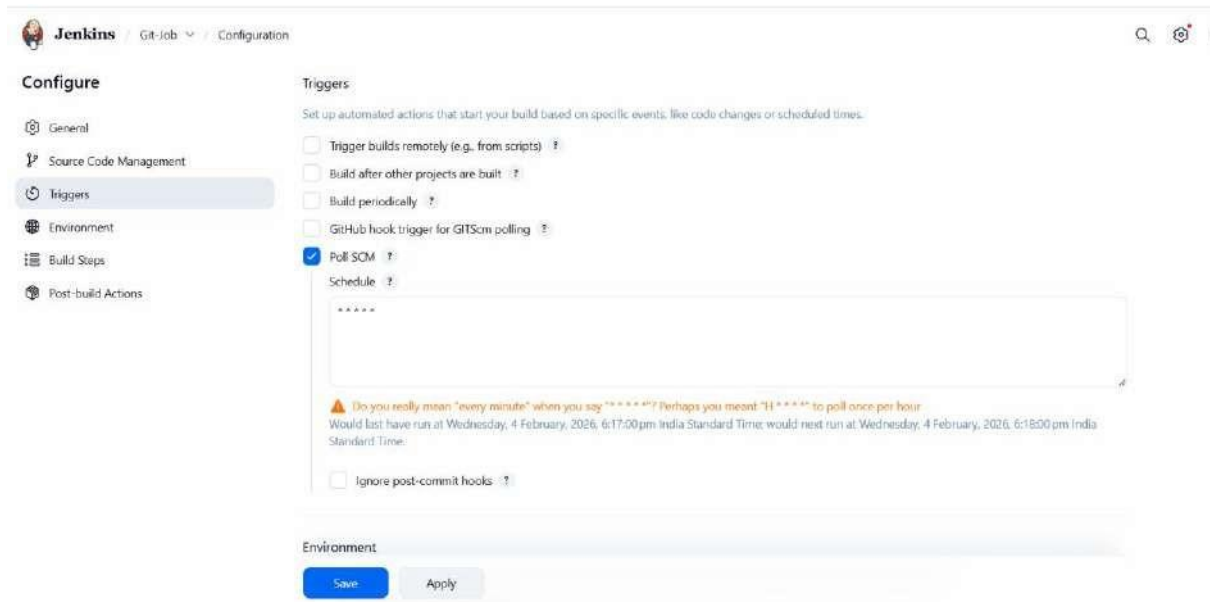
04-02-2026 18:13 <DIR>      .
04-02-2026 18:10 <DIR>      ..
04-02-2026 18:13          27 hello.txt
04-02-2026 18:13          16 README.md
                2 File(s)      41 bytes
                2 Dir(s)  67,768,750,080 bytes free

C:\Users\likit\.jenkins\workspace\Git-Job>exit 0
Finished: SUCCESS
```

REST API Jenkins 2.528.3

Poll SCM / Webhook trigger

Build automatically on code change



Automatic build on code change



Jenkins / Git-Job



Status



Changes



Workspace



Build Now



Configure



Delete Project



Git Polling Log



Rename

Builds



Filter



Today



#3 6:20 PM



#2 6:13 PM



localhost:8080/job/Git-Job/3/console

Jenkins Git-Job #3 Console Output

Status

Changes

Console Output

Edit Build Information

Delete build #3

Polling Log

Timings

Git Build Data

Previous Build

Download Copy View as plain text

Started by an SCM change
Running as SYSTEM
Building in workspace C:\Users\likit\.jenkins\workspace\Git-Job
The recommended git tool is: NONE
No credentials specified
> C:\Program Files\Git\bin\git.exe rev-parse --resolve-git-dir C:\Users\likit\.jenkins\workspace\Git-Job\git # timeout=10
Fetching changes from the remote Git repository
> C:\Program Files\Git\bin\git.exe config remote.origin.url https://github.com/likit-git/jenkins-demo.git # timeout=10
Fetching upstream changes from https://github.com/likit-git/jenkins-demo.git
> C:\Program Files\Git\bin\git.exe --version # timeout=10
> git --version # 'git version 2.52.0.windows.1'
> C:\Program Files\Git\bin\git.exe fetch --tags --force --progress -- https://github.com/likit-git/jenkins-demo.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> C:\Program Files\Git\bin\git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision 7d5d331e1a12e3c5c9ccca30fefea7e8a0546c0f (refs/remotes/origin/main)
> C:\Program Files\Git\bin\git.exe config core.sparsecheckout # timeout=10
> C:\Program Files\Git\bin\git.exe checkout -f 7d5d331e1a12e3c5c9ccca30fefea7e8a0546c0f # timeout=10
Commit message: "Update hello.txt with new content"
> C:\Program Files\Git\bin\git.exe rev-list --no-walk 410f4dc29973be777003f1e1e85673b8e2b95e8f # timeout=10
[Git-Job] \$ cmd /c call C:\Users\likit\AppData\Local\Temp\jenkins4167640230753574945.bat

C:\Users\likit\.jenkins\workspace\Git-Job>echo code pulled successfully
code pulled successfully

localhost:8080/job/Git-Job/3/console

Jenkins Git-Job #3 Console Output

> C:\Program Files\Git\bin\git.exe checkout -f 7d5d331e1a12e3c5c9ccca30fefea7e8a0546c0f # timeout=10
Commit message: "Update hello.txt with new content"
> C:\Program Files\Git\bin\git.exe rev-list --no-walk 410f4dc29973be777003f1e1e85673b8e2b95e8f # timeout=10
[Git-Job] \$ cmd /c call C:\Users\likit\AppData\Local\Temp\jenkins4167640230753574945.bat

C:\Users\likit\.jenkins\workspace\Git-Job>echo Code pulled successfully
Code pulled successfully

C:\Users\likit\.jenkins\workspace\Git-Job>dir
Volume in drive C is Windows-SSD
Volume Serial Number is AC5E-ED6E

Directory of C:\Users\likit\.jenkins\workspace\Git-Job

04-02-2026 18:120 <DIR> .
04-02-2026 18:110 <DIR> ..
04-02-2026 18:120 48 hello.txt
04-02-2026 18:113 14 README.md
2 File(s) 62 bytes
2 Dir(s) 67,765,092,352 bytes free

C:\Users\likit\.jenkins\workspace\Git-Job>exit 0
Finished: SUCCESS

localhost:8080

REST API Jenkins 2.528.3

Jenkins

+ New Item

Build History

Build Queue

No builds in the queue.

Build Executor Status

0/2

All +

S	W	Name	Last Success	Last Failure	Last Duration
✓	☀	first-job	55 min #1	N/A	0.8 sec
✓	☁	Git-Job	8 min 33 sec #3	17 min #1	3.9 sec

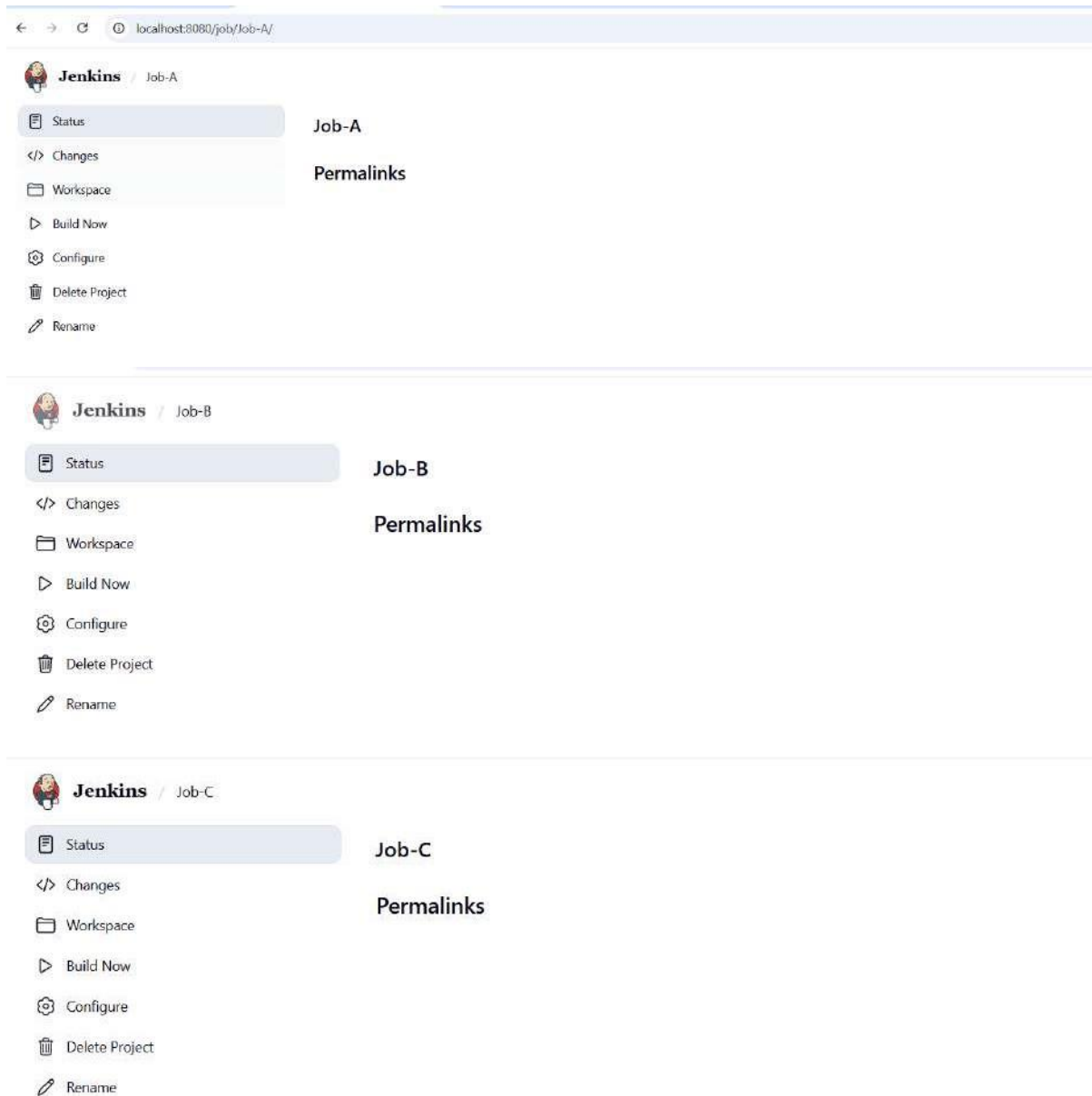
Icons: S M L

III. Learning Goal

Understand job flow visually

Create multiple Freestyle jobs:

- Job-A (Build)
- Job-B (Test)
- Job-C (Deploy)



Use:

- Post-build action → Trigger other projects

Jenkins

Job-A

Configuration

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

+ Add build step

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

Build other projects

Projects to build

Job-B

Trigger only if build is stable

Trigger even if the build is unstable

Trigger even if the build fails

+ Add post-build action

Save

Apply

localhost:8080/job/Job-B/

Jenkins

Job-B

Status

Changes

Workspace

Build Now

Configure

Delete Project

Rename

Job-B

Upstream Projects

Job-A

Permalinks

Builds

Jenkins

Job-B

Configuration

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

+ Add build step

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

Build other projects

Projects to build

Job-C

Trigger only if build is stable

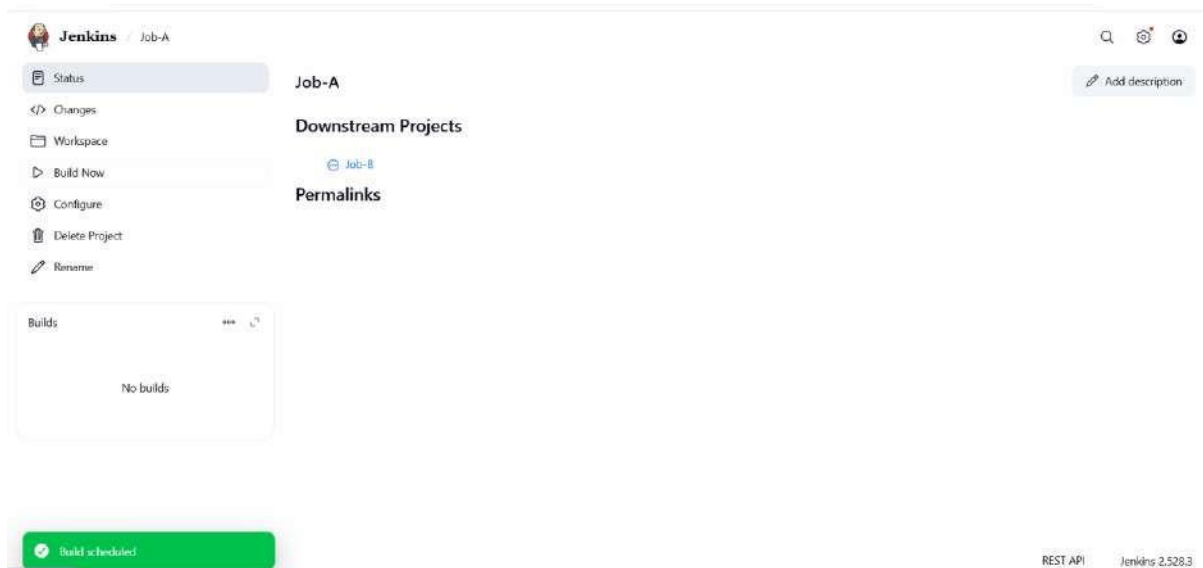
Trigger even if the build is unstable

Trigger even if the build fails

+ Add post-build action

Save

Apply

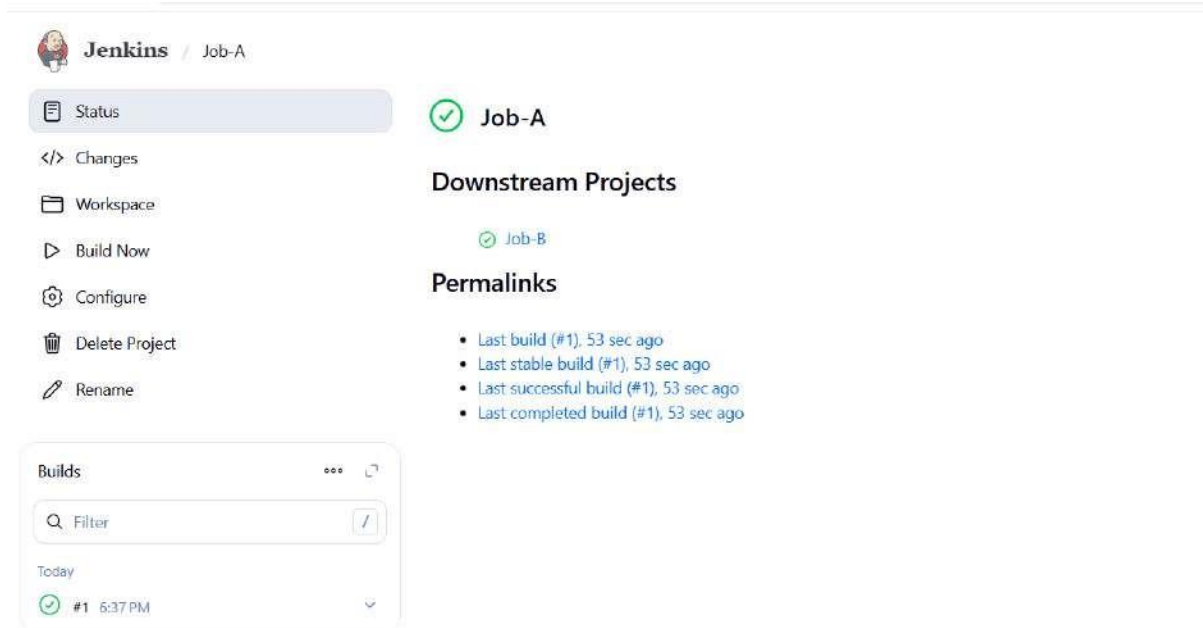


The Jenkins Job-A overview page shows a sidebar with navigation links: Status, Changes, Workspace, Build Now, Configure, Delete Project, and Rename. The main content area displays 'Job-A' with a green status icon, 'Downstream Projects' (Job-B), and 'Permalinks'. A 'Builds' section shows 'No builds'. A green banner at the bottom indicates 'Build scheduled'. The REST API and Jenkins version (2.528.3) are shown in the bottom right.

View job chaining visually



The Jenkins Job-A Console Output page shows the build log for Job-A. The log text is: 'Started by user P. Lixitha', 'Running as SYSTEM', 'Building in workspace C:\Users\likit\.jenkins\workspace\Job-A', '[Job-A] \$ cmd /c call C:\Users\likit\AppData\Local\Temp\jenkins13251940043910366127.bat', 'c:\Users\likit\.jenkins\workspace\Job-A>echo Building the project', 'Building the project', 'c:\Users\likit\.jenkins\workspace\Job-A>exit @', 'Triggering a new build of Job-B', 'Finished: SUCCESS'. The page includes a sidebar with navigation links and buttons for Download, Copy, and View as plain text.



The Jenkins Job-A overview page shows a sidebar with navigation links: Status, Changes, Workspace, Build Now, Configure, Delete Project, and Rename. The main content area displays 'Job-A' with a green status icon, 'Downstream Projects' (Job-B), and 'Permalinks'. A 'Builds' section shows a list of builds: '#1 6:37 PM'. The 'Permalinks' section lists: 'Last build (#1), 53 sec ago', 'Last stable build (#1), 53 sec ago', 'Last successful build (#1), 53 sec ago', and 'Last completed build (#1), 53 sec ago'.

 Jenkins / Job-B

Status

Changes

Workspace

Build Now

Configure

Delete Project

Rename

Builds

Filter

Today

#1 6:37 PM

Job-B

Upstream Projects

Job-A

Downstream Projects

Job-C

Permalinks

- Last build (#1), 2 min 0 sec ago
- Last stable build (#1), 2 min 0 sec ago
- Last successful build (#1), 2 min 0 sec ago
- Last completed build (#1), 2 min 0 sec ago

 Jenkins / Job-C

Status

Changes

Workspace

Build Now

Configure

Delete Project

Rename

Builds

Filter

Today

#1 6:37 PM

Job-C

Upstream Projects

Job-B

Permalinks

- Last build (#1), 2 min 9 sec ago
- Last stable build (#1), 2 min 9 sec ago
- Last successful build (#1), 2 min 9 sec ago
- Last completed build (#1), 2 min 9 sec ago

 Jenkins

+ New Item

Build History

Build Queue

No builds in the queue.

Build Executor Status

0/2

All

5

	W	Name	Last Success	Last Failure	Last Duration	
✓	☀	first-job	1 hr 17 min #1	N/A	0.8 sec	▶
✓	☁	Git-Job	29 min #3	39 min #1	3.9 sec	▶
✓	☀	Job-A	12 min #1	N/A	0.6 sec	▶
✓	☀	Job-B	12 min #1	N/A	0.26 sec	▶
✓	☀	Job-C	12 min #1	N/A	0.17 sec	▶

🔍

⚙️

👤

Add description

Install **Pipeline Stage View**

localhost:8080/manage/pluginManager/available

Jenkins / Manage Jenkins / Plugins

Plugins

Search: Pipeline Stage View

Install Available plugins Installed plugins Advanced settings Download progress

Install	Name	Released	Health
☐	Pipeline: Stage View 2.39 User interface Pipeline Stage View Plugin.	23 days ago	100

localhost:8080/manage/pluginManager/updates/

Jenkins / Manage Jenkins / Plugins

Plugins

Download progress

Preparation

- Checking internet connectivity
- Checking update center connectivity
- Success

commons-lang3 v3.x Jenkins API	Success
Icons API	Success
Folders	Success
OWASP Markup Formatter	Success
ASAP API	Success
JSON Path API	Success
Structs	Success
Pipeline: Step API	Success
commons-text API	Success
Token Macro	Success
Build Timeout	Success
bouncycastle API	Success
Credentials	Success
Plain Credentials	Success
Variant	Success
SSH Credentials	Success
Credentials Binding	Success
SCM API	Success

localhost:8080

Jenkins / Pipeline-Job / Configuration

Configure

General Triggers Pipeline Advanced

Poll SCM

Trigger builds remotely (e.g., from scripts)

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

```
1 pipeline {
2   agent any
3   stages {
4     stage('Build') {
5       steps {
6         echo 'Building project'
7       }
8     }
9     stage('Test') {
10      steps {
11        echo 'Running tests'
12      }
13    }
14  }
15 }
```

try sample Pipeline...

Save Apply

Jenkins / Pipeline-Job / Configuration

Configure

- General
- Triggers
- Pipeline
- Advanced

Definition

Pipeline script:

```
Script ?
7   }
8   }
9   stage('Test') {
10  steps {
11    echo 'Running tests'
12  }
13  }
14  stage('Deploy') {
15  steps {
16    echo 'Deploying app'
17  }
18  }
19 }
20 }
21 }
```

☒ Use Groovy Sandbox

Pipeline Syntax

Save Apply

Observe stage-wise execution

localhost:8080/job/Pipeline-Job/

Jenkins / Pipeline-Job

- Status
- Changes
- Build Now
- Configure
- Delete Pipeline
- Full Stage View
- Stages
- Rename
- Pipeline Syntax

Pipeline-Job

Stage View

Average stage times:
(full run time: ~5s)

	Build	Test	Deploy
Feb 04 18:52 No Changes	237ms	137ms	126ms

Permalinks

Builds

Today

✓ #1 6:52 PM

The image displays two screenshots of the Jenkins web interface, specifically the 'Console Output' view for a pipeline job named 'Pipeline-Job' (build #1).

Top Screenshot: Shows the initial execution of the pipeline. The console output includes the following steps:

- Started by user p.likitha
- [Pipeline] start of Pipeline
- [Pipeline] node
- Running on Jenkins in C:\Users\likitha\jenkins\workspace\Pipeline-Job
- [Pipeline] {
- [Pipeline] stage
- [Pipeline] { (Build)
- [Pipeline] echo
- Building project
- [Pipeline] }
- [Pipeline] // stage
- [Pipeline] stage
- [Pipeline] { (Test)
- [Pipeline] echo
- Running tests
- [Pipeline] }
- [Pipeline] // stage
- [Pipeline] stage
- [Pipeline] { (Deploy)
- [Pipeline] echo
- Deploying app
- [Pipeline] }
- [Pipeline] // stage
- [Pipeline] }
- [Pipeline] // node

Bottom Screenshot: Shows the final state of the pipeline execution. The console output includes the following steps:

- [Pipeline] {
- [Pipeline] stage
- [Pipeline] { (Build)
- [Pipeline] echo
- building project
- [Pipeline] }
- [Pipeline] // stage
- [Pipeline] stage
- [Pipeline] { (Test)
- [Pipeline] echo
- Running tests
- [Pipeline] }
- [Pipeline] // stage
- [Pipeline] stage
- [Pipeline] { (Deploy)
- [Pipeline] echo
- Deploying app
- [Pipeline] }
- [Pipeline] // stage
- [Pipeline] }
- [Pipeline] // node
- [Pipeline] End of Pipeline
- Finished: SUCCESS

Both screenshots show a left-hand sidebar with navigation options: Status, Changes, Console Output (selected), Edit Build Information, Delete build '#1', Timings, Pipeline Overview, Restart from Stage, Replay, Pipeline Steps, and Workspaces. The top right of the interface includes search, settings, and user profile icons, along with buttons for Download, Copy, and View as plain text.

IV. Learning Goal

Use Jenkins Pipeline & Jenkinsfile

1.What is Jenkins Pipeline?

A **Jenkins Pipeline** is a way to define the **entire build, test, and deployment process as code**.

Instead of clicking steps manually in Jenkins UI, we write a script that tells Jenkins:

- how to get code
- how to build it
- how to test it
- how to deploy it

This script is usually stored in a file called **Jenkinsfile** inside the project repository.

Key Idea:

Pipeline = **Automation workflow written as code**

Benefits:

- Automates CI/CD process
- Version-controlled (stored in Git)
- Repeatable and reliable
- Shows stages visually (Build → Test → Deploy)

2. Scripted vs Declarative Pipeline

Feature	Scripted Pipeline	Declarative Pipeline
Style	Code-based (Groovy scripting)	Structured, simple syntax
Difficulty	Advanced	Easy to learn
Flexibility	Very flexible	Less flexible but safer
Syntax	Uses node {} and Groovy code	Uses pipeline {} block
Error chances	Higher	Lower

Feature	Scripted Pipeline	Declarative Pipeline
Used by	Experts, complex logic	Most projects

Eg:

Scripted Pipeline

```
node {  
    stage('Build') {  
        echo 'Building'  
    }  
    stage('Test') {  
        echo 'Testing'  
    }  
}
```

Declarative Pipeline

```
pipeline {  
    agent any  
    stages {  
        stage('Build') {  
            steps {  
                echo 'Building'  
            }  
        }  
        stage('Test') {  
            steps {  
                echo 'Testing'  
            }  
        }  
    }  
}
```



```

    }
}
}
}

```

```

1 pipeline {
2   agent any
3   stages {
4     stage('Build') {
5       steps {
6         echo 'Build Stage from Jenkinsfile'
7       }
8     }
9     stage('Test') {
10      steps {
11        echo 'Test Stage from Jenkinsfile'
12      }
13    }
14    stage('Deploy') {
15      steps {
16        echo 'Deploy Stage from Jenkinsfile'
17      }
18    }
19  }
20 }
21

```

Commit changes

Commit message
Add Jenkinsfile with build, test, and deploy stages

Extended description
Add an optional extended description...

Message and description suggested by Copilot

☒ Commit directly to the main branch

☐ Create a new branch for this commit and start a pull request [Learn more about pull requests](#)

Cancel Commit changes

V. Learning Goal

Pipeline as code from repository

Create **Pipeline Job**

Create Jenkinsfile

 Jenkins

All ▾ / New Item

New Item

Enter an item name:

Jenkinsfile-Pipeline

Select an item type

 Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

 Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

 Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

 Folder

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

 Multibranch Pipeline

Builds and deploys multiple branches of a project automatically.

OK

Commit Jenkinsfile to GitHub

 Jenkins

Jenkinsfile-Pipeline ▾ / Configuration

Configure

 General

 Triggers

 Pipeline

 Advanced

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script from SCM ▾

SCM ⓘ

Git ▾ ⓘ

Repositories ⓘ

Repository URL ⓘ

https://github.com/Luli-git/jenkins-demo.git

Credentials ⓘ

- none - ▾ + Add

Advanced ▾

Jenkins

Jenkinsfile-Pipeline

Configuration

Search

Help

Profile

Configure

General

Triggers

Pipeline

Advanced

Branch Specifier (blank for 'any')

*/main

+ Add Branch

Repository browser

(Auto)

Additional Behaviours

+ Add

Script Path

Jenkinsfile

☒ Lightweight checkout

Pipeline Syntax

Save

Apply



Jenkins

Jenkinsfile-Pipeline

#1

Console Output

Status

Changes

Console Output

Edit Build Information

Delete build #1

Timings

Git Build Data

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

Workspaces

Console Output

Download

Copy

View as plain text

Started by user P Likitha
Obtained Jenkinsfile from git https://github.com/Liki-git/jenkins-demo.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\Users\likitha\jenkins\workspace\Jenkinsfile-Pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: checkout scm)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/Liki-git/jenkins-demo.git
> C:\Program Files\Git\bin\git.exe init C:\Users\likitha\jenkins\workspace\Jenkinsfile-Pipeline # timeout=10
Fetching upstream changes from https://github.com/Liki-git/jenkins-demo.git
> C:\Program Files\Git\bin\git.exe --version # timeout=10
> git --version # 'git version 2.52.0.windows.1'
> C:\Program Files\Git\bin\git.exe fetch --tags --progress -- https://github.com/Liki-git/jenkins-demo.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> C:\Program Files\Git\bin\git.exe config remote.origin.url https://github.com/Liki-git/jenkins-demo.git # timeout=10
> C:\Program Files\Git\bin\git.exe config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> C:\Program Files\Git\bin\git.exe rev-parse --refs/remotes/origin/main # timeout=10
Checking out Revision 949ac521e5da2db62fdcf5fc236c9498cec51421 (refs/remotes/origin/main)

lockhost8388

JenkinsJenkinsfile-Pipeline#1Console Output

```
Avoid second fetch
> C:\Program Files\Git\bin\git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision 949ac521e5da2db62fdcf5fc236c94986c51421 (refs/remotes/origin/main)
> C:\Program Files\Git\bin\git.exe config core.sparsecheckout # timeout=10
> C:\Program Files\Git\bin\git.exe checkout -f 949ac521e5da2db62fdcf5fc236c94986c51421 # timeout=10
Commit message: "Add Jenkinsfile with build, test, and deploy stages"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
Build Stage from Jenkinsfile
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Test Stage from Jenkinsfile
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploy Stage from Jenkinsfile
[Pipeline] }
```

JenkinsJenkinsfile-Pipeline#1Console Output

```
[Pipeline] { (Build)
[Pipeline] echo
Build Stage from Jenkinsfile
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Test Stage from Jenkinsfile
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploy Stage from Jenkinsfile
[Pipeline] }
[Pipeline] // stage
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] end of Pipeline
Finished: SUCCESS
```

REST API

Jenkins 2.528.3

Jenkins

+ New Item

Build History

Build Queue

No builds in the queue

Build Executor Status

0/2

All

+

S	W	Name	Last Success	Last Failure	Last Duration	
✓	☀	first-job	1 hr 29 min #1	N/A	0.8 sec	▶
✓	☁	Git-Job	4 min 52 sec #4	51 min #1	3.5 sec	▶
✓	☀	Jenkinsfile-Pipeline	2 min 20 sec #1	N/A	7.7 sec	▶
✓	☀	Job-A	24 min #1	N/A	0.6 sec	▶
✓	☀	Job-B	24 min #1	N/A	0.26 sec	▶
✓	☀	Job-C	24 min #1	N/A	0.17 sec	▶
✓	☀	Pipeline-Job	9 min 25 sec #1	N/A	5.9 sec	▶

Icons: S M L

Add description